Next-Step Experimental Plan

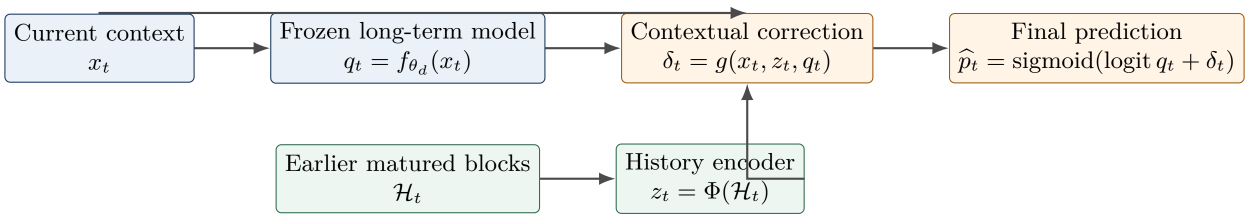
Learning Causal Within-Day Signals for CTR Prediction

A capacity ladder from neural sequence encoders to linear adapters

CTR-Predictions Project • September 2026

Purpose. The previous experiment showed that a single online calibration intercept has almost no room to improve Criteo or Avazu, while full Online Platt Scaling yields only a very small gain. This does *not* rule out useful within-day information: heterogeneous campaign- or context-level changes can cancel in a global scalar statistic. The next experiment will test whether a more expressive but still causal and lightweight history encoder can extract such information.

Core design. Freeze the existing long-term predictor within each day. Convert matured observations from earlier time blocks into a common causal history representation. Feed exactly that representation to a ladder of models, from a context-query Transformer down to a linear interaction adapter. Select the *simplest* model that produces a reproducible gain over the long-only predictor and Online Platt Scaling.



1 Research Questions and Testable Hypotheses

The experiment is designed to answer four questions.

1. **Is there predictive information in the causal within-day stream beyond the long-term model and clock time?**
2. **Is that information heterogeneous across campaigns or contexts, so that a global calibrator cannot capture it?**
3. **How much function-approximation capacity is required: sequence attention, recurrence, a fixed-window MLP, bilinear interactions, or only a linear model?**
4. **Can the useful part be retained in a computationally efficient and theoretically controlled form?**

The hypotheses are:

- **H1 (incremental signal):** at least one contextual history model improves test log loss over both the long-only predictor and Online Platt Scaling.
- **H2 (context interaction):** a model that lets the current impression query the historical state outperforms a context-free history encoder.
- **H3 (chronology):** at least part of the improvement disappears when the order of within-day history blocks is shuffled.
- **H4 (parsimony):** if a simpler model is statistically tied with a more complex model, the simpler model is preferred.

2 Common Causal Prediction Setup

For impression i on day d, let the existing long-term system output

$$q_{d,i} = f_{\theta_d}(x_{d,i}), \tag{1}$$

where θ_d and all long-term mixture weights are fixed throughout day d. Clip $q_{d,i}$ to $[10^{-5}, 1-10^{-5}]$ before taking logits.

The within-day method produces a residual logit correction:

$$\widehat{p}_{d,i} = \mathrm{sigmoid}\big(\mathrm{logit}(q_{d,i}) + \delta_{d,i}\big)\,. \tag{2}$$

The output layer is initialized at zero, so $\delta_{d,i} = 0$ initially and the model exactly recovers the long-only predictor.

2.1 Causal blocks and feedback delay

Partition each day into fixed blocks. The primary configuration is:

$$\text{block width} = 15 \text{ minutes}, \qquad \text{feedback delay} = 30 \text{ minutes}. \tag{3}$$

A block may enter the history only after the block is complete and its assumed feedback delay has elapsed. Sensitivity checks use 5- and 30-minute blocks and delays of 0 and 60 minutes, but only after the architecture is selected on validation data.

Define the base residual for a matured impression j as

$$r_j = y_j - q_j. \tag{4}$$

Reset the within-day history state at the beginning of each day in the main specification. Carrying the previous day's state is an ablation, not the default.

2.2 A common block token for all variants

To make the capacity comparison fair, every variant receives the same information. Let $c(x) \in \mathbb{R}^m$ be a fixed signed-hash sketch of the ten categorical fields, normalized to have bounded norm. Use $m \in \{32, 64\}$, selected on validation data.

For completed block k, construct

$$\begin{aligned} u_k = \big[&\log(1+n_k),\ \overline{y}_k,\ \overline{q}_k,\ \overline{\mathrm{logit}(q)}_k,\ \overline{r}_k,\ \overline{|r|}_k, \\ &\overline{\ell(q,y)}_k,\ \overline{c(x)}_k,\ \overline{r\,c(x)}_k\big]. \end{aligned} \tag{5}$$

The residual-weighted sketch $\overline{r\,c(x)}_k$ is the key heterogeneous-drift feature: it records which regions of context space were recently underpredicted or overpredicted. All components are computed only from matured impressions.

The current-impression input is

$$a_{d,i} = \big[c(x_{d,i}),\ \mathrm{logit}(q_{d,i}),\ \text{time-of-day}_{d,i}\big]. \tag{6}$$

Time of day is supplied to all learned variants and to a dedicated baseline, so a gain cannot be attributed merely to regular daily seasonality.

2.3 Leakage prevention

- Base predictions used to train the adapter must be prequential or otherwise out-of-sample. Never train on in-sample residuals from the base model.
- For impression i, the history contains only blocks that are available before its prediction time under the declared feedback delay.
- Hyperparameters and architecture selection use development days only. The locked test is opened once after the selection rule is frozen.
- The base predictor and long-term mixture are identical across all variants.

3 Capacity Ladder: Complex to Simple

All variants use Equation (2), the same block tokens in Equation (5), and the same current-impression input. Their only substantive difference is the function class used for $\delta_{d,i}$.

3.1 V1: Context-query Transformer

Use the most recent $K = 16$ available block tokens, corresponding to four hours under the primary block width. Add relative-age and time-of-day encodings. The current impression creates the query:

$$z_{d,i} = \mathrm{Attention}\left(Q = W_Q a_{d,i},\ K = W_K U_{d,i},\ V = W_V U_{d,i}\right), \tag{7}$$

Variant	Function approximator	Question answered
V1	Context-query Transformer	Can current context retrieve the relevant part of the recent block sequence?
V2	Contextual GRU adapter	Is a compact streaming recurrent state sufficient?
V3	Fixed-window contextual MLP	Is recurrence unnecessary once recent levels and trends are explicit?
V4	Low-rank bilinear adapter	Are simple context-history interactions sufficient?
V5	Linear interaction adapter	Can a sparse, interpretable, theory-friendly model capture the signal?

where $U_{d,i}$ is the sequence of available block tokens. Then

$$\delta_{d,i} = \delta_{\max} \tanh(\mathrm{MLP}[a_{d,i}, z_{d,i}]) \, . \tag{8}$$

Use one Transformer layer, two heads, and hidden dimension 32 as the default. This model is the flexible upper end of the ladder, not necessarily the final preferred method.

3.2 V2: Contextual GRU adapter

Process each completed block once:

$$h_k = \mathrm{GRU}(h_{k-1}, u_k). \tag{9}$$

For the current impression, use

$$\delta_{d,i} = \delta_{\max} \tanh\big(\mathrm{MLP}[a_{d,i}, h_k, a'_{d,i} \odot W h_k]\big) \, . \tag{10}$$

The interaction term is essential: without it, the GRU can collapse to a nonlinear global calibrator. Use one layer and hidden dimension 32 by default.

3.3 V3: Fixed-window contextual MLP

Construct a deterministic summary

$$s_{d,i} = \big[u_k, u_{k-1}, \mathrm{EWMA}_1(u), \mathrm{EWMA}_4(u), \mathrm{EWMA}_{16}(u), u_k - u_{k-1}\big], \tag{11}$$

where the subscripts denote block-scale half-lives. Predict

$$\delta_{d,i} = \delta_{\max} \tanh(\mathrm{MLP}[a_{d,i}, s_{d,i}]) \, . \tag{12}$$

Use two hidden layers of widths 64 and 32. This model tests whether explicit recent levels and trends are enough, without recurrence or attention.

3.4 V4: Low-rank bilinear adapter

Use the same fixed summary $s_{d,i}$ and model

$$\delta_{d,i} = \alpha^\top s_{d,i} + \beta^\top a_{d,i} + \left(U^\top a_{d,i}\right)^\top \left(V^\top s_{d,i}\right) , \tag{13}$$

with rank 4 or 8. This is the most important middle-ground model: it allows the same historical state to affect different current contexts differently while remaining small and interpretable.

3.5 V5: Linear interaction adapter

Create bounded features

$$\phi_{d,i} = \big[1, a_{d,i}, s_{d,i}, \mathrm{hash}(a_{d,i} \otimes s_{d,i})\big] \tag{14}$$

and predict

$$\delta_{d,i} = w^\top \phi_{d,i}. \tag{15}$$

Use both an offline regularized logistic fit and an online projected-gradient or FTRL version. The hashed interaction block is required; without it, this model is little more than global calibration.

3.6 Theory-friendly frozen-encoder variant

For V1–V3, also retain the encoder output as a fixed feature after offline training and update only a linear residual head online:

$$\widehat{p}_t = \mathrm{sigmoid}\big(\mathrm{logit}(q_t) + w_t^\top \phi_t\big)\,, \qquad w_{t+1} = \Pi_{\|w\|\le B}\left[w_t - \eta_t(\widehat{p}_t - y_t)\phi_t\right]. \tag{16}$$

If $\|\phi_t\| \le R$ and $\eta = B/(R\sqrt{T})$, standard online gradient descent gives

$$\sum_{t=1}^{T} \ell_t(w_t) - \min_{\|w\|\le B} \sum_{t=1}^{T} \ell_t(w) \le BR\sqrt{T}. \tag{17}$$

Because $w = 0$ recovers the long-only predictor, Equation (17) also provides a direct safeguard relative to the base model up to the regret term. The feature sequence may be adaptive and history-dependent, provided ϕ_t is formed before observing y_t.

4 Baselines and Diagnostic Controls

Method	Correction	Role
Long-only	$\delta_t = 0$	Existing strongest long-term predictor; primary reference.
Time-of-day	Fixed hour/block intercepts learned from prior days	Tests regular seasonality without current-day labels.
Online intercept	$\delta_t = b_t$	Reproduces the negative scalar-calibration result.
Online Platt	$\widehat{p}_t = \mathrm{sigmoid}(a_t \,\mathrm{logit}\, q_t + b_t)$	Strong standard online-calibration baseline.
Campaign calibrator	$\delta_t = b_{0,t} + b_{\mathrm{campaign}(x_t),t}$ with shrinkage	Simple heterogeneous-drift baseline.
Context-only control	Same architecture as V1–V3, but all history tokens are zero	Controls for added model capacity.
Shuffled-history control	Shuffle prior block order within day	Tests whether chronology, rather than daily composition, matters.

An optional robustness comparison applies Online Platt Scaling to the plain expanding-history backbone. This tests whether a weak long-term model plus calibration can match the proposed combination, but it is not a substitute for comparisons using the same long-term backbone.

5 Training and Model Selection Protocol

5.1 Stage A: build one reusable causal replay cache

For each impression, cache:

- timestamp, day, block, label, and current categorical context;
- frozen long-term prediction q_t;
- the set of matured prior blocks available at prediction time;
- block tokens u_k and deterministic summaries s_t;
- fixed context sketch $c(x_t)$.

All variants must read the same cache. This is the main reproducibility safeguard.

5.2 Stage B: train adapters without changing the base predictor

Use chronological adapter-training days, followed by separate development days. Minimize log loss with optional penalties

$$\lambda_\delta \mathbb{E}[\delta_t^2] + \lambda_2 \|\psi\|_2^2. \tag{18}$$

Initialize the final correction layer to zero, use early stopping on development log loss, and cap corrections with $\delta_{\max} \in \{1, 2\}$. Do not optimize AUC directly.

Use a deliberately small validation grid:

Context-sketch dimension	$32, 64$
Hidden dimension	$16, 32, 64$ where applicable
Learning rate	$3 \times 10^{-4}, 10^{-3}$
Weight decay	$10^{-5}, 10^{-4}$
Dropout	$0, 0.1$
Bilinear rank	$4, 8$
Correction cap $\delta_{\max}$	$1, 2$

Keep block width, feedback delay, and history construction fixed during architecture selection. Tune temporal granularity only after choosing the winning function class.

5.3 Stage C: frozen test protocol

Preserve the current chronological splits and locked-test discipline. Use the same seed set for all methods. At minimum, retain the current three-seed Criteo and eight-seed Avazu protocol; use additional seeds only if declared before inspecting test results.

For each test day:

1. load the frozen long-term model and adapter parameters;
2. reset the within-day state;
3. replay impressions and mature blocks causally;
4. store every prediction before updating any state with its label;
5. report both predictive metrics and computational cost.

6 Evaluation Metrics and Statistical Analysis

6.1 Primary and secondary metrics

- **Primary:** impression-weighted log loss over the locked test.
- **Robustness:** mean, median, and worst daily log loss; fraction of days beating long-only and Online Platt.
- **Calibration:** Brier score, expected calibration error, and residual curves over time of day.
- **Ranking:** AUC and average precision. These become relevant because contextual corrections can change ranking, unlike a global intercept.
- **Temporal localization:** log loss in early, middle, and late portions of each day and in blocks with high base-model residual drift.
- **Efficiency:** training time, state-update time, serving time per million impressions, peak memory, and parameter count.

Use paired daily differences as the unit of statistical analysis. Report paired bootstrap confidence intervals over days, sign tests, and mean $\pm$ standard deviation across random seeds. Do not use impression-level independence assumptions for significance claims.

6.2 Required headline comparisons

For every candidate V1–V5, report

$$\Delta_{\text{long}} = L(\text{candidate}) - L(\text{long-only}), \qquad \Delta_{\text{OPS}} = L(\text{candidate}) - L(\text{Online Platt}). \tag{19}$$

Also report the chronology contribution

$$\Delta_{\text{chron}} = L(\text{shuffled history}) - L(\text{real history}), \tag{20}$$

and the context-interaction contribution

$$\Delta_{\text{ctx}} = L(\text{context-free}) - L(\text{contextual}). \tag{21}$$

7 Ablation Plan

Run the following in priority order after selecting the strongest development-set architecture.

1. **No history:** zero all history tokens while retaining the same network and time-of-day input.
2. **Shuffled chronology:** permute prior block order within each day.
3. **No context interaction:** history affects all impressions through the same correction.
4. **No residual-weighted sketch:** remove $\overline{r\,c(x)}_k$.
5. **Label-free history:** retain traffic and context-composition summaries but remove y, residual, and loss features.
6. **Campaign-only history:** replace the hashed contextual sketch with campaign-level summaries.
7. **Temporal granularity:** compare 5-, 15-, and 30-minute blocks and delays of 0, 30, and 60 minutes.
8. **Reset versus carryover:** reset each morning versus decay the previous day's terminal state.

The first five ablations are required. The remaining three are conditional on a positive development result.

8 Decision Rules

Advance a within-day model to the locked test only if all of the following hold on development days:

1. it beats long-only and Online Platt in mean log loss;
2. the gain is stable across seeds and is not driven by one day;
3. it beats its same-capacity no-history control;
4. either $\Delta_{\text{chron}} > 0$ or $\Delta_{\text{ctx}} > 0$ by a practically visible margin;
5. its serving cost remains compatible with the project goal of a lightweight add-on.

Model selection rule. Among models satisfying the above, select the simplest model whose validation loss is within one standard error of the best model. Do not select the Transformer merely because it has the lowest point estimate.

Before opening the test set, declare a materiality floor for absolute log-loss improvement; a reasonable starting point is 2×10^{-4} on Criteo, subject to confirmation from development-day variability. A statistically detectable but operationally negligible improvement should be reported as such.

Proceed to downstream autobidding only if the selected model beats both long-only and Online Platt on the locked prediction test and the gain is not reproduced by shuffled or time-only controls. Otherwise, retain Online Platt as a baseline and stop the within-day branch on these datasets.

9 Execution Order and Deliverables

1. **Causal cache and reproduction:** reproduce long-only, online intercept, Online Platt, and time-of-day results from one shared cache.
2. **Complex models:** train V1 Transformer and V2 GRU with fixed default sizes.
3. **Intermediate models:** train V3 MLP and V4 bilinear adapter.
4. **Simple models:** train V5 offline linear and V5 online linear/FTRL.
5. **Development selection:** apply the predeclared parsimony rule.
6. **Core ablations:** no history, shuffled chronology, no context interaction, no residual-weighted sketch, and label-free history.
7. **Locked test:** evaluate the single selected architecture, its essential baselines, and the required controls.
8. **Conditional extensions:** temporal-granularity sensitivity, frozen-encoder online head, and downstream autobidding.

The implementation should produce:

- one row-level prediction file per method and seed;
- one daily/block-level metrics table;
- a parameter-count and latency table;
- plots of cumulative loss difference, time-of-day residuals, and gain versus drift intensity;
- a short decision memo stating whether within-day information is genuinely useful, whether contextual interaction is necessary, and which level of model capacity is justified.

10 Expected Interpretation of Outcomes

Observed result	**Interpretation and next action**
V1/V2 beat OPS; simple models do not	Useful nonlinear or sequential within-day structure exists; retain the smallest successful neural encoder and test the frozen-encoder online head.
V3/V4 match V1/V2	Recurrence and attention are unnecessary; prefer the MLP or bilinear model.
V5 matches neural models	Use the linear adapter as the final method and develop the online regret result.
Models beat no-history but not shuffled history	Gain comes from daily composition or static context, not chronology; do not claim streaming adaptation.
Contextual models beat global models	Global cancellation explains the failure of scalar calibration; heterogeneous drift is the central mechanism.
No model beats OPS and controls	Within-day history is not meaningfully exploitable in these public datasets; refocus on cross-day adaptive historical learning.

The experiment is designed to produce a useful conclusion even if all complex models fail: it will distinguish a lack of signal from a lack of function-approximation capacity and prevent further investment in unsupported within-day mechanisms.